

Exercise Sheet: Application Layer

Exercise 1

- 1.1: What are sockets used for?
- 1.2: What does HTTP stand for and what is the protocol used for?
- 1.3: Is the following statement true for HTTP?
“HTTP is stateless by default, but statefulness can be enabled by the client”
 Give a brief explanation of your answer.
- 1.4: What is the difference between persistent and non-persistent HTTP?
- 1.5: What is the difference between the two HTTP status codes 403 and 404?
- 1.6: Why was HTTP/2 introduced?
- 1.7: What is the purpose of web caches?
- 1.8: Is it true that SMTP is used to send, receive, and access emails? Give a brief explanation of your answer.
- 1.9: What is the difference between the approaches to trusting keys in OpenPGP and S/MIME?

Exercise 2

- 2.1: Explain the term “Client-Server Paradigm”.
- 2.2: Explain the term “Peer-to-peer Architecture”.
- 2.3: Explain the term “elastic” in the context of network throughput.
- 2.4: Explain the difference between iterated and recursive queries in DNS.

Exercise 3

In this exercise you will make first experiences with capturing and analysing network data.

3.0: Download & install Wireshark on your laptop/PC from: <https://www.wireshark.org/download.html>. Then, familiarize yourself with the Wireshark UI and how to capture live network data using chapters 3&4 of its documentation: https://www.wireshark.org/docs/wsug_html_chunked/index.html

3.1: Start a live network data capture. While running the live capture, visit the website <http://example.com> in your browser. Then wait a few moments and reload the page in your browser. Then stop the capture. By looking at the captured data, answer the following questions:

- a) How long did it take for the server to respond to the first HTTP GET with an HTTP OK?
- b) What is the IP address of your computer?
- c) What is the IP address of the example.com server?

- d) Is your browser running HTTP version 1.0, 1.1, or 2? What version of HTTP is the server running?
- e) What is the HTTP status code returned by the server in the first HTTP GET request?
- f) How many bytes of content are being returned to your browser in the response to the first HTTP GET request?
- g) How long is the data that was sent to the browser as response to the first HTTP GET request valid if it is cached? Which header field tells you this info?
- h) How does the HTTP GET request to the server change when reloading the page? How does the server's response change from the original response?
- i) Find the first DNS query involving the domain example.com. To which IP address was the query sent?
- j) What type is specified in the DNS query?
- k) What is the answer data in the response to the query?